

Prevalence of stenotic changes in arteries supplying the lumbar spine. A postmortem angiographic study on 140 subjects

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OBJECTIVES—To study the prevalence of arterial diseases in the arteries supplying the lumbar spine and their relation to other vascular diseases, as well as to chronic low back pain.

METHODS—Five pairs of the lumbar arteries and the middle sacral artery were evaluated from 140 postmortem aortograms, performed in connection with routine medicolegal necropsies on subjects ranging from 16 to 89 years of age. For information about low back pain history, a close relative of each of the deceased was interviewed two to four weeks after the necropsy.

RESULTS—Twenty one (22%) men and nine (20%) women had occluded arteries, and an additional 33 (35%) men and 17 (38%) women had narrowed arteries. The mean age for men with occluded or narrowed arteries, or both, was 50 years and for women 59 years. Most of the stenotic changes were seen at the orifices or in the first part of the arteries. The middle sacral artery was most often affected, followed by the fourth lumbar arteries. The number of collateral arteries increased with occluded ($p < 0.001$) and narrowed arteries ($p = 0.001$). Stenotic lumbar/middle sacral arteries were found, on average, five years earlier than atherosclerosis of the coronary arteries. Subjects with one or more occluded/narrowed arteries were 8.5 times more likely to have suffered from chronic (that is, three months or longer) low back pain at some time during their life than were those without such findings (odds ratio adjusted for age and sex 8.5; 95% confidence intervals 2.9, 24; $p < 0.001$).

CONCLUSIONS—The study shows that the lumbar and middle sacral arteries frequently become obliterated by atheromatous lesions during adult life, and that obliteration of these arteries is more common in subjects with a history of chronic back pain than in those without.